5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

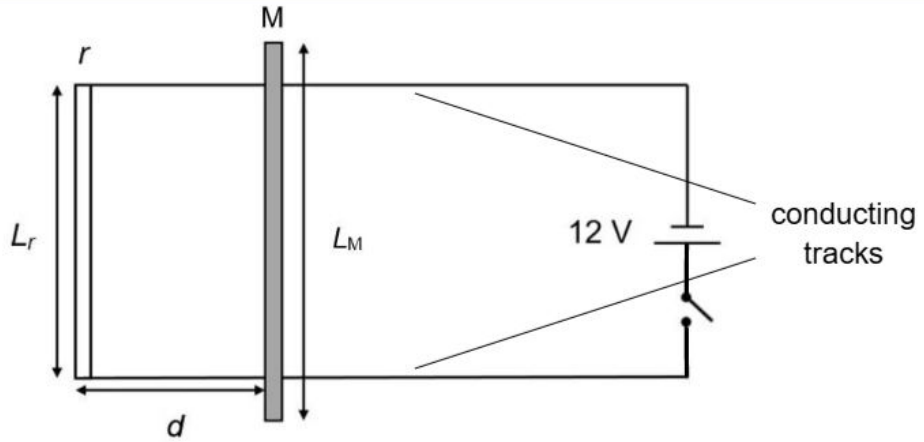


Fig. 5.1 (not to scale)

When the switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

the current passing through r, $I_r = \dfrac{12\text{ V}}{6\ \Omega} = 2.0\text{ A}$

$B = \dfrac{\mu_0 I_r}{2\pi d} = \dfrac{(4\pi \times 10^{-7}) \times 2.0}{2\pi(0.040)} = 1.0 \times 10^{-5}\text{ T} = 10\ \mu\text{T}$ (shown) [B1]

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

$I_M = \dfrac{12\text{ V}}{(\frac{0.50}{0.60})(12\ \Omega)} = 1.2\text{ A}$

The magnetic force on M, acting over the current-carrying length of 0.50 m within the field of r:

$F = BI_M L = (1.0 \times 10^{-5})(1.2)(0.50)$ [M1]

$= 6.0 \times 10^{-6}$ N [A1]

Direction of F: leftwards, from Fleming's left-hand rule. [B1]

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.
Describe and explain the initial motion of M.

M moves and accelerates to the right. [B1]

The **applied magnetic field of 1.0 mT is much larger than 10 μT produced by r**. By Fleming's left-hand rule, the **applied field exerts a force to the right on M, which is larger than the 6.0 × 10^{-6} N force to the left due to the field of** *r*. Hence, the net force on M is to the right. [B1]

..... [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.
The proton follows the path shown in Fig. 5.2.

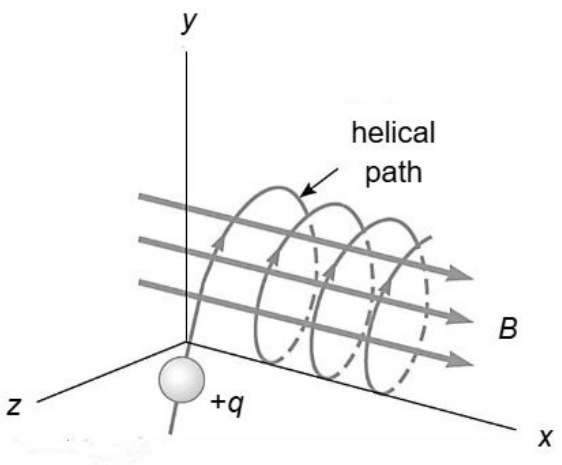


Fig. 5.2

(i) Explain why the proton follows this helical path.

Since the proton enters at an angle to *B*, its velocity has two components: **one perpendicular to *B* and one parallel to *B***. [B1]

The perpendicular component **experiences a magnetic force** ($F = Bqv\sin\theta$) that is always **perpendicular to its velocity**, **providing the centripetal force** for uniform circular motion. The parallel component experiences **no magnetic force** and so the proton continues to move along the direction of *B* at **constant speed**. The combination of these two motions produces a helical path. [B1]

.....

.....

..... [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed v at the position shown.

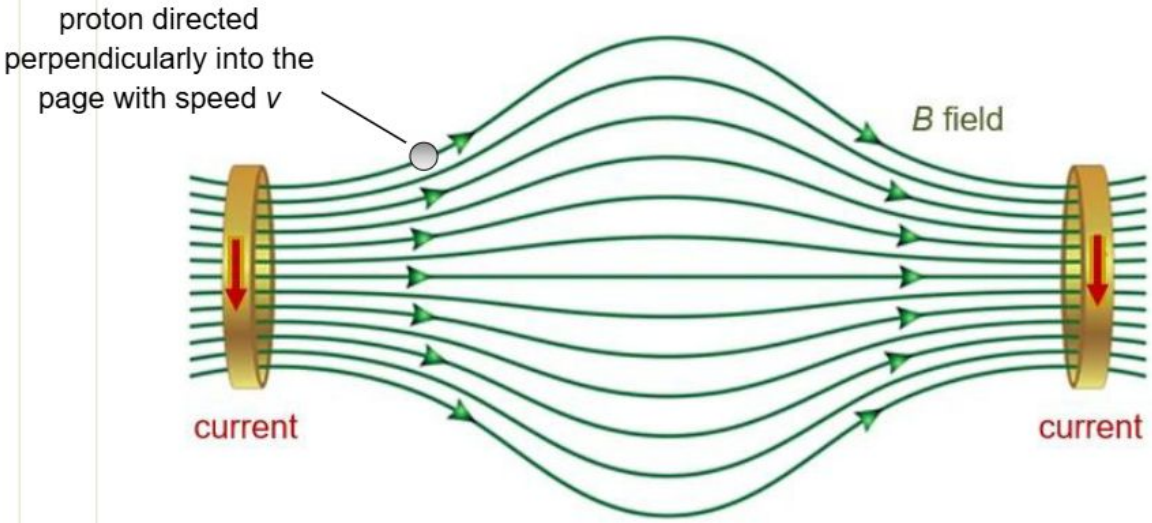


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

The proton will experience a magnetic force perpendicular to the magnetic field based on Flemming's left hand rule. This **magnetic force, pointing towards the centre of the magnetic bottle**, produces velocity parallel and perpendicular to the magnetic field. **Past the centre** of the magnetic bottle, the magnetic force perpendicular to the field, **pointing towards the centre of the magnetic bottle**, opposes the velocity parallel to magnetic field, (reducing its magnitude and changes the direction of velocity eventually) [B1*]

This combined motion of a velocity parallel and perpendicular to the magnetic field in the non-uniform magnetic field produces a **spiral motion** from **one end of the magnetic bottle to the other** where the **radius of the motion increases towards the middle and decreases towards the ends**.[B1]

This process continues and the proton oscillate indefinitely between the two coils, while spiralling around the axis of the bottle.

[To take note] Throughout the motion, since the magnetic force is always perpendicular to the velocity, it does no work on the proton and the speed of the proton remains constant — only the distribution between the velocity components parallel and perpendicular to the B field such that the kinetic energy remains constant.

[Total: 10]